Residual Spatial Attention Kernel Generation Network for Hyperspectral Image Classification With Small Sample Size

Yanbing Xu, Yanmei Zhang, Chengcheng Yu, Chao Ji, Tingxuan Yue, and Huan Li

***Abstract*— With the rapid development of deep learning, the convolutional neural networks (CNNs) have been widely used in hyperspectral image classification (HSIC) and achieved excellent performance. However, CNNs reuse the same kernel weights over different locations, thus resulting in the insufficient capability of capturing diversity spatial interactions. Moreover, CNNs usually require a large amount of training samples to optimize the learnable parameters. When training samples are limited, the classification performance of CNN tends to drop off a cliff. To tackle the aforementioned issues, a novel residual spatial attention kernel generation network (RSAKGN) is proposed for HSIC. First, a spatial attention kernel generation module (SAKGM) is built to extract discriminative semantic features, which can dynamically calculate the attention weights to generate specific spatial attention kernels over different locations. Then, we combine the SAKGM with residual learning framework by embedding the SAKGM into a bottleneck residual block to obtain the residual spatial attention block (RSAB). The RSAKGN is constructed by stacking several RSABs. Experimental results on three public HSI datasets demonstrate that the proposed RSAKGN method outperforms several state-of-the-arts with small sample size.**

***Index Terms*— Attention mechanism, hyperspectral image classification (HSIC), residual learning framework, small sample size, spatial attention kernel generation module (SAKGM).**

I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spectral information and complex spatial correlation, which can simultaneously describe the physical characteristics and spatial distribution of ground objects. This enables HSIs to be widely used in a variety of fields, such as urban development [1], environment management [2], and geological exploration [3]. Many excellent research works have emerged in HSI analysis and processing fields (e.g., classification [4], target detection [5], endmember extraction [6], and data fusion [7]). For instance, Hong *et al.* [4] developed a novel mini-batch graph convolutional network (GCN), which allows training large-scale GCNs in a mini-batch fashion and effectively improves classification performance.

HSI classification (HSIC) task attempts to identify the category for each hyperspectral pixel [8]–[10], which is the cornerstone of many hyperspectral remote sensing applications. Traditional HSIC methods usually include two steps: feature engineering and classifier design. The HSI features are first manually extracted according to prior knowledge [11]–[13] and then sent to the vanilla classifier for classification [14]–[18]. However, the traditional method has the disadvantages of poor generalization ability and insufficient representation capacity, thus resulting in the inability to extract more discriminative deep semantic features.

In the past few years, deep-learning-based methods have been widely applied in the classification of HSIs and have achieved excellent performance. The stacked autoencoder (SAE) [19] and the deep belief network (DBN) [20] were first introduced for the HSIC. However, these methods require flattening the 3-D HSI patch cube to a 1-D vector as the input data of the network, which will destroy the spatial context information and the intrinsic interaction in the 3-D HSI patch cube. Many 2-D convolutional neural network (CNN) methods [21]–[24] were proposed to address this issue, which can directly process 3-D HSI patch cube and extract the high-level semantic features. For example, Chen *et al.* [19] proposed a siamese CNN consisting of two CNNs that share weight parameters and supervised by a margin ranking loss function. To further explore the spatial–spectral information in the 3-D HSI patch, 3-D CNNs [25]–[27] were proposed to simultaneously extract deep spatial–spectral features.

However, with the deepening of the network, the model will have many problems, such as gradient disappearance, gradient explosion, and overfitting, which will lead to the decline of the classification accuracy. To address the above issues, some works introduced the deep residual learning framework [28] into their models for the HSIC, such as [29]–[31]. Song *et al.* [29] proposed a deep feature-fusion network with residual connection, which can extract discriminative multilevel features via a very deep architecture and fuse mechanism. Paoletti *et al.* [30] proposed a deep pyramidal residual network for spectral–spatial HSIC, which can gradually widen the convolutional layers and increase the diversity of the deep sematic features as the network deepens.

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Although these CNN-based methods have achieved excellent performance in HSIC, they require a large amount of training samples to optimize the learnable parameters; the available training samples are expensive to be obtained. Once the training samples are limited, the CNN-based methods may provide disappointing classification results. Li *et al.* [32] proposed a novel pixel-pair method to significantly increase the training samples without any additional labeled samples and adopted a voting strategy to determine the final pixel label. Zheng *et al.* [33] proposed a novel sample enlargement method by performing superpixel segmentation on the raw HSI data, which can effectively enhance the training set and achieve excellent classification results. Zhong *et al.* [34] designed a spectral–spatial residual network (SSRN) with two sequential residual 3-D convolution blocks, which can consecutively learn discriminative features and achieve excellent performance on a small training dataset. Duan *et al.* [35] combined manifold learning with sparse representation to develop a new semisupervised feature extraction (FE) algorithm called geodesic-based sparse manifold hypergraph (GSMH), which can achieve satisfactory FE performance with limited labeled training samples. Gao *et al.* [36] combined residual learning framework with depthwise convolution, which not only deepens the network but also effectively reduces network parameters, and achieved satisfactory results with limited training samples. Dong *et al.* [37] proposed a novel pixel cluster CNN and spectral–spatial fusion algorithm, which utilize the CNN framework to extract effective features from the pixel clusters. Although these methods achieve satisfactory results by enhancing the training sample sets or reducing the training parameters, they rarely inherently improve the capability of modeling diverse spatial correlations with small sample size.

Recently, some works [38]–[44] introduced the attention mechanism to model the intrinsic interactions in the HSI data. For example, Zhu *et al.* [38] proposed a residual spectral–spatial attention network (RSSAN), which can adaptively enhance the meaningful spectral–spatial information and suppress useless information in a pixel-centered neighborhood. Sun *et al.* [39] combined the 3-D CNN with spatial attention mechanism to design an spectral–spatial attention network (SSAN), which can capture discriminative spectral–spatial features from the HSI cubes. Zhong *et al.* [43] proposed a spectral–spatial transformer network (SSTN), which performed spatial attention and spectral association on the raw HSI data to model the spectral–spatial interactions in a 3-D HSI cube. Hong *et al.* [44] rethought HSIC from a sequential perspective with transformers and proposed a novel backbone network called SpectralFormer to mine the sequence attributes of spectral signatures. Although these attention-based methods provided satisfactory classification results, they only focused on the global feature spaces, ignoring the diversity of the intrinsic interaction in a pixel-centered neighborhood over different positions, which limiting the further improvement for HSIC.

To conquer the aforementioned problems, this article proposed a residual spatial attention kernel generation network (RSAKGN) for HSIC. Specifically, we proposed a spatial-attention-based dynamic kernel generation mechanism, which could fully utilize the particular context information to calculate the attention weights over different spatial locations. In this way, the RSAKGN can utilize those diverse spatial attention kernels to extract discriminative semantic features effectively. In addition, we combine the spatial attention kernel generation module (SAKGM) with residual learning framework to construct a residual spatial attention block (RSAB), which can further improve the representation capacity of deep semantic features. Then, the RSAKGN is constructed by stacking several RSABs.

The main contribution of this article is as follows.

1) A novel SAKGM is proposed to capture the diverse spatial interactions, which can dynamically generate spatial attention kernels according to the particular context information over different positions. The SAKGM is a plug and play module which can be flexibly embedded into networks to improve classification performance.
2) An RSAB is proposed to extract discriminative semantic features from HSI data. The RSAB is constructed by embedding SAKGMs into a residual bottleneck block to replace the convolutional layers, which can not only improve the classification accuracy but also reduce the number of training parameters.
3) An RSAKGN is proposed for small sample size HSIC. The RSAKGN is constructed by stacking several RSABs, which can extract multilevel semantic features effectively without optimization by large amounts of training samples.
4) Extensive experiments are conducted on three public HSI datasets to verify the effectiveness of the RSAKGN. The experimental results demonstrate that the proposed RSAKGN outperforms other state-of-the-arts and can cope with small sample size problems.

The remainder of this article is organized as follows. First, the related works to HSIC are reviewed in Section II. Then, a detailed description of the proposed RSAKGN is given in Section III. Next, the experimental results and evaluations of the proposed method are reported in Section IV. Finally, the final conclusion and discussion of future research works are presented in Section V.

II. Related Work

A. Involution Operation

The involution operation [45] is proposed to model the diverse intrinsic interactions over different positions and has attracted widespread attention in computer vision. Recently, Meng *et al.* [46] combined the involution operation with residual learning to construct a deep residual involution network (DRIN) for HSIC, which could adaptively allocate weights over different spatial positions according to the spectral information of target pixel. Although the DRIN achieved excellent performance in HSIC, the involution kernels only explicitly focus on the target spectral vectors, thus sacrificing the pixel-paired correlation within an HSI cube. And the generation of involution kernels still relies on the feature mapping performed by fully connected layers, which requires a large amount of training samples to optimize the learnable parameters. As a result, the DRIN cannot cope with small

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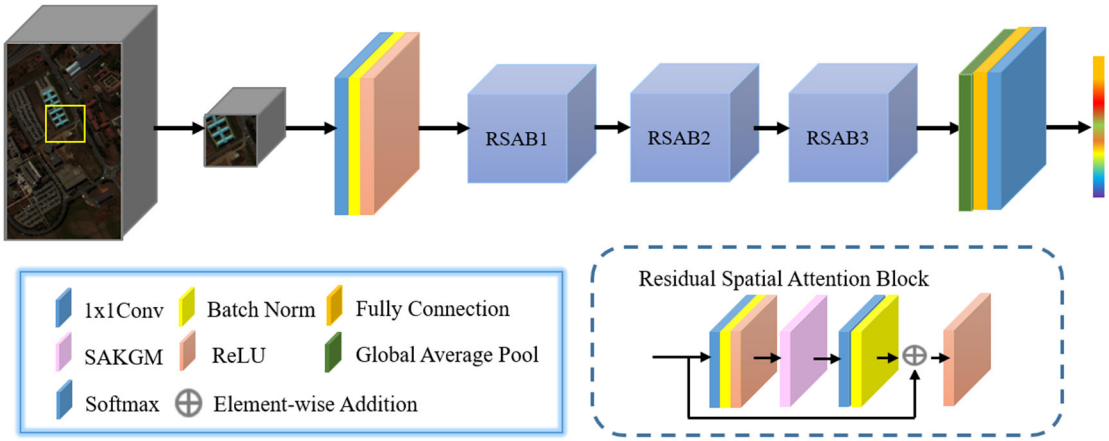


Fig. 1. Framework of the proposed RSAKGN for HSIC.

sample size problems. Different from the DRIN, a novel attention-based kernel generation module which can fully exploit the diverse spatial context to extract discriminative semantic features is proposed in this article. The proposed attention-based kernel generation module exploits attention mechanism, which is capable for modeling the diverse pixel-paired correlation within an HSI cube, instead of fully connected layers to generate dynamic spatial attention kernels for the FE. The proposed module can be flexibly embedded into CNNs to extract discriminative semantic features effectively for HSIC with small sample size.

III. Methodology

A. Overview of the Proposed Model

First, we define the variables related to HSIC. Let $\boldsymbol{H} \in R^{H\times W\times C}$ represent the HSI data, where H, W, and C denote the spatial height and width of the HSI and the number of spectral bands, respectively. The HSI data can also be denoted as $\boldsymbol{H} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{HW}\}$, where $\boldsymbol{x}_i \in R^{1\times 1\times C}$ is the pixel vector in HSI data. In order to make full use of the spatial and spectral information of the HSI, we take the pixel $\boldsymbol{x}_i$ in HSI as the center pixel and generate a 3-D HSI cube set $\boldsymbol{\chi} = \{\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_{HW}\}$, where $\boldsymbol{X}_i \in R^{S\times S\times C}$, $S \times S$ denotes the patch size of HSI cubes, and each HSI cube $\boldsymbol{X}_i$ corresponds to the pixel $\boldsymbol{x}_i$. The category label of HSI cube $\boldsymbol{X}_i$ is determined by the center pixel $\boldsymbol{x}_i$. Then, the category label set that corresponding to the HSI cube set $\boldsymbol{\chi}$ is defined as $\mathrm{Y} = \{\boldsymbol{Y}_1, \boldsymbol{Y}_2, \ldots, \boldsymbol{Y}_{HW}\}$, where $\boldsymbol{Y}_i \in \{1, 2, \ldots, n\}$ and n denotes the number of land-cover categories. The sample dataset can be represented as $\mathrm{Z}(\boldsymbol{\chi}, \mathrm{Y})$. Then, the dataset Z is randomly divided into training set $\mathrm{Z}_{\text{train}}$, validation set Z_{val}, and test set Z_{test} according to a certain proportion.

Fig. 1 shows the framework of the proposed model which consists of a 1×1 convolutional layer, three RSABs, and a fully connected classification layer. In order to make full use of the spatial and spectral information simultaneously, the raw 3-D HSI cubes are extracted as the input of RSAKGN.

B. Spatial Attention Kernel Generation Module

Different from the traditional convolution operations which reuse static convolution kernel weights to extract spatial features over different positions, the SAKGM can fully utilize the context information to dynamically generate specific spatial attention kernels over different positions. These generated kernels can effectively extract discriminative deep semantic features without requiring a large amount of training samples for parameter optimization, which enables the model competent for small sample size HSIC. Fig. 2 shows the structure of the proposed SAKGM.

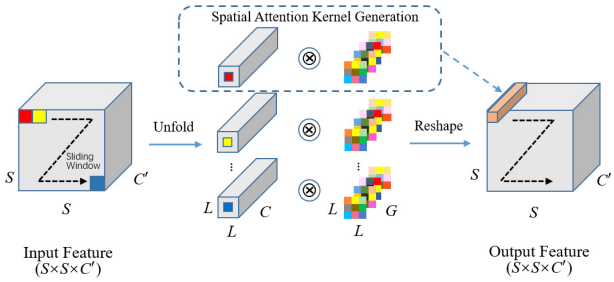


Fig. 2. Structure of the proposed SAKGM.

1) Unfold Operation: Define $\boldsymbol{H}_i \in R^{S\times S\times C'}$ to represent the semantic feature of the HSI cube $\boldsymbol{X}_i$, where $S\times S$ is the spatial size and C' is the channel number. For the input feature $\boldsymbol{H}_i = \{\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_{\text{ss}}\}$, unfold operation is performed on it to obtain a pixel-centered neighborhood set $\mathfrak{D} = \{\boldsymbol{D}_1, \boldsymbol{D}_2, \ldots, \boldsymbol{D}_{\text{ss}}\}$, where $\boldsymbol{D}_j \in R^{L\times L\times C'}$, $L \times L$ is the spatial size of the spatial attention kernels, and $\boldsymbol{h}_j$ is the center pixel of the corresponding pixel neighborhood $\boldsymbol{D}_j$. Specifically, the unfold operation is to resample $\boldsymbol{H}_i$ by sliding windows where the spatial size of sliding window is set to $L \times L$ and the stride is set to 1. The zero-padding strategy is adopted in the resampling process. The process of unfold operation is shown in Fig. 3.

2) Spatial Attention Kernel Generation Mechanism: Spatial attention mechanism aims to model the interactions between different locations of HSI samples [41], [42]. In this article, we use a variant of the spatial attention mechanism to calculate the spatial attention kernel over different locations.

For the pixel-centered neighborhood set $\mathfrak{D} = \{\boldsymbol{D}_1, \boldsymbol{D}_2 \cdots \boldsymbol{D}_{\text{ss}}\}$ which generated by the unfold operation, we aim to generate a spatial attention kernel set

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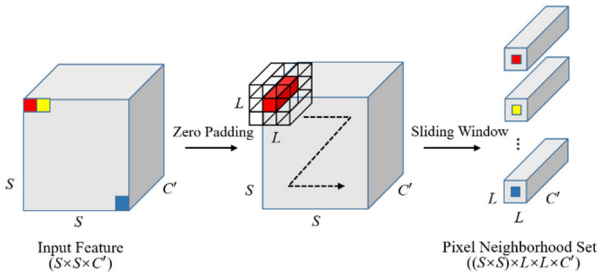


Fig. 3. Illustration of the unfold operation.

$\boldsymbol{\Phi} = \{\boldsymbol{W}_1, \boldsymbol{W}_2, \ldots, \boldsymbol{W}_{ss}\}$, where the spatial attention kernel $\boldsymbol{W}_j \in R^{L\times L\times G}$ corresponds to the pixel-centered neighborhood $\boldsymbol{D}_j$ one-to-one. We take a single pixel neighborhood $\boldsymbol{D}_j$ as an example to elaborate the spatial attention kernel generation mechanism. Fig. 4 demonstrates the calculation flow of the spatial attention kernel generation mechanism.

Different from the ordinary spatial attention mechanism which calculates the affinity matrix between each pixel-paired within an HSI cube, the proposed algorithm focus on the correlations between the feature representation of $\boldsymbol{D}_j$ and pixels within $\boldsymbol{D}_j$. Therefore, it is crucial to seek a compact and accurate feature representation of $\boldsymbol{D}_j$ as the query tensor $\boldsymbol{q}_j$ for the subsequent affinity matrix calculation.

First, to generate the global average eigenvector $\boldsymbol{d}_j^{\text{avg}}$ which can compactly represent the summary statistics of $\boldsymbol{D}_j$, the global average pooling (GAP) is operated on each spectral feature across spatial dimensions $L \times L$, as done in [38] and [47]. The $\boldsymbol{d}_j^{\text{avg}}$ can be formulated as follows:

$$\boldsymbol{d}_j^{\text{avg}} = \frac{1}{L\times L}\sum_{k=1}^{L}\sum_{l=1}^{L}\boldsymbol{d}(j,k) \in R^{1\times 1\times C'} \tag{1}$$

where $\boldsymbol{d}(k,l)$ denotes the pixel vector at position (k,l) in $\boldsymbol{D}_i$. In addition, the convolutional block attention module (CBAM) [48] indicates that the global max pooling also has complementary global information for GAP. The global maximum eigenvector $\boldsymbol{d}_j^{\max}$ is also calculated for the global feature representation, and the $\boldsymbol{d}_j^{\max}$ is calculated by

$$\boldsymbol{d}_j^{\max} = \max\left(\boldsymbol{D}_j\right) \in R^{1\times 1\times C'} \tag{2}$$

where the function $\max(\cdot)$ denotes the global max pooling. To make the feature representation of $\boldsymbol{D}_j$ more discriminative, the center pixel vector $\boldsymbol{h}_j$, which belonging to the raw input feature $\boldsymbol{H}_i$ and containing abundant discriminative spectral signatures, is selected to calculate the pixel-paired relationships within $\boldsymbol{D}_j$, as done in ordinary spatial attention mechanism. The local identity eigenvector $\boldsymbol{d}_j^{\text{ident}}$ can be formulated as follows:

$$\boldsymbol{d}_j^{\text{ident}} = \boldsymbol{h}_j \in R^{1\times 1\times C'}. \tag{3}$$

Then, all the three eigenvectors are added up to obtain a compact and comprehensive feature representation of $\boldsymbol{D}_j$. The query tensor $\boldsymbol{q}_j$ of the spatial attention mechanism can be formulated as follows:

$$\boldsymbol{q}_j = \boldsymbol{d}_j^{\text{avg}} + \boldsymbol{d}_j^{\max} + \boldsymbol{d}_j^{\text{ident}} \in R^{1\times 1\times C'}. \tag{4}$$

The previous work [38] has demonstrated that the multihead attention mechanism is helpful to capture diverse semantic features. Inspired by the multihead attention mechanism, we use 1×1 convolutional layer to map the query vector $\boldsymbol{q}_j$ into a high-dimensional feature space $\boldsymbol{Q}_j$ for the subsequent generation of multihead spatial attention kernels. The $\boldsymbol{Q}_j$ can be formulated as follows:

$$\boldsymbol{Q}_j = \mathbb{F}\left(\boldsymbol{W}_Q, \boldsymbol{q}_j\right) \in R^{1\times 1\times (GC')} \tag{5}$$

where $\mathbb{F}(\cdot)$ denotes a 2-D convolutional operation with kernel size of $(GC') \times C' \times 1 \times 1$, $\boldsymbol{W}_Q \in R^{(GC')\times C'}$ is the trainable parameters of $\mathbb{F}(\cdot)$, and G denotes the number of spatial attention kernels. Then, the $\boldsymbol{Q}_j$ and $\boldsymbol{D}_j$ are reshaped to the size of $G \times C'$ and $LL \times C'$ for the following matrix multiplication. The correlation maps $\boldsymbol{A}_j$ between $\boldsymbol{D}_j$ and $\boldsymbol{Q}_j$ can be computed as follows:

$$\boldsymbol{A}_j = \left(\frac{\boldsymbol{D}_j}{\|\boldsymbol{D}_j\|_2}\right)\left(\frac{\boldsymbol{Q}_j}{\|\boldsymbol{Q}_j\|_2}\right)^{\mathrm{T}} \in R^{LL\times G} \tag{6}$$

where $\|\cdot\|_2$ denotes the L_2-norm along the last dimension of input tensor. The spatial attention kernel $\boldsymbol{W}_j$, which corresponding to the pixel-centered neighborhood $\boldsymbol{D}_j$, can be calculated as follows:

$$\boldsymbol{W}_j = \text{sfm}_{-1}\left(\boldsymbol{A}_j\right) \in R^{LL\times G} \tag{7}$$

where $\text{sfm}_{-1}(\cdot)$ denotes the softmax function along the first dimension of the input tensor. Then, we reshape $\boldsymbol{W}_j$ to the size of $L \times L \times G$ and divide $\boldsymbol{D}_j$ equally into G groups along the channel dimension for the following group element-wise multiplication and summation. Fig. 5 demonstrates the calculation flow of the group element-wise multiplication and summation. The mth channel's element value of the output tensor $\boldsymbol{out}_j$ can be calculated as follows:

$$\text{out}_j^m = \sum_{u=1}^{L}\sum_{v=1}^{L} w_j\left(u, v, \lceil mG/C'\rceil\right) d_j(u,v,m) \tag{8}$$

where m denotes the channel index, $\lceil\cdot\rceil$ denotes the rounded up function, and $w_j(u,v,\lceil mG/C'\rceil)$ and $d_j(u,v,m)$ denote the element value at position $(u,v,\lceil mG/C'\rceil)$ in $\boldsymbol{W}_j$ and the element value at position (u,v,m) in $\boldsymbol{D}_i$, respectively.

C. Residual Spatial Attention Block

Residual learning framework [29] has been proven to be effective in solving gradient vanishing and gradient exploding problems, and has been widely used in computer vision, such as SENet [47] and CBAM [48]. Some previous works [30]–[32] also introduced the residual learning framework into HSIC and achieved good classification performance. Therefore, we also embed the proposed SAKGM into the ordinary bottleneck residual block to obtain an RSAB. Fig. 6 shows the structure of the RSAB and the ordinary bottleneck residual block.

The RSAB consists of two 1×1 convolutional layers and one SAKGM. The first 1×1 convolutional layer is used to reduce the channel numbers of the input feature $\boldsymbol{H}^{\text{input}}$ to decrease the computational complexity of the SAKGM. Then,

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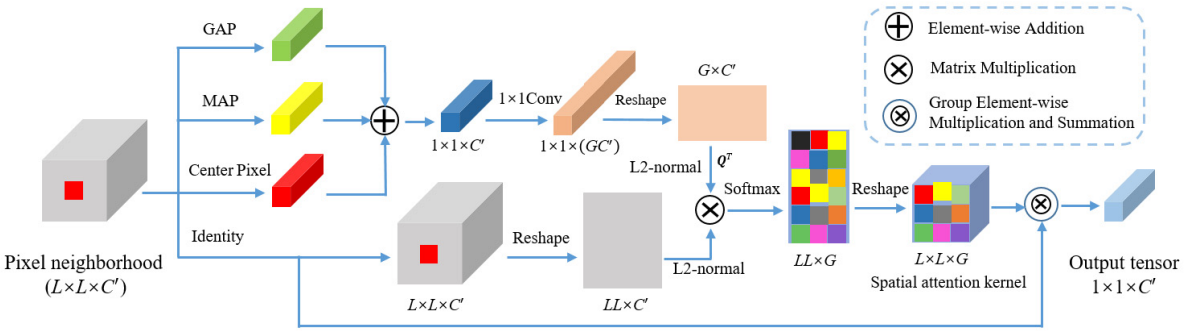


Fig. 4. Illustration of the spatial attention kernel generation mechanism used in SAKGM.

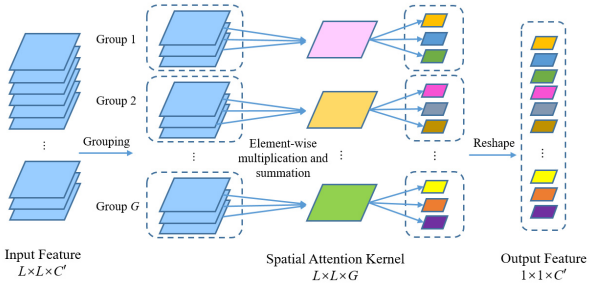


Fig. 5. Illustration of the group element-wise multiplication and summation.

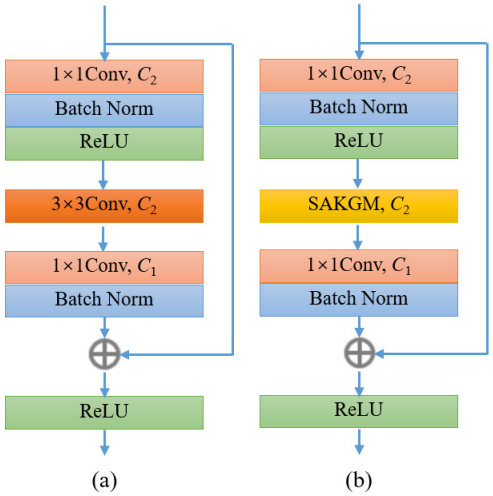


Fig. 6. Structure of: (a) bottleneck residual block and (b) RSAB.

the SAKGM is responsible for the extraction of deep semantic features $\boldsymbol{H}^{\text{SAKGM}}$. The second 1×1 convolutional layer is used to map the semantic feature $\boldsymbol{H}^{\text{SAKGM}}$ into a new feature space $\boldsymbol{H}^{\text{restore}}$ which has the same channel numbers as the original input feature $\boldsymbol{H}^{\text{input}}$. Finally, the original input feature $\boldsymbol{H}^{\text{input}}$ is added to $\boldsymbol{H}^{\text{restore}}$ through skip connections to generate the final output. The final output $\boldsymbol{H}^{\text{output}}$ can be formulated as follows:

$$\boldsymbol{H}^{\text{reduce}} = \delta\left(\Theta\left(\mathcal{F}_1\left(\boldsymbol{W}^{\text{reduce}}, \boldsymbol{H}^{\text{input}}\right)\right)\right) \tag{9}$$

$$\boldsymbol{H}^{\text{SAKGM}} = \psi\left(\boldsymbol{H}^{\text{reduce}}\right) \tag{10}$$

$$\boldsymbol{H}^{\text{restore}} = \mathcal{F}_2\left(\boldsymbol{W}^{\text{restore}}, \boldsymbol{H}^{\text{SAKGM}}\right) \tag{11}$$

$$\boldsymbol{H}^{\text{output}} = \delta\left(\Theta\left(\boldsymbol{H}^{\text{restore}} + \boldsymbol{H}^{\text{input}}\right)\right) \tag{12}$$

where $\mathcal{F}_1(\cdot)$ and $\mathcal{F}_2(\cdot)$ denote the 2-D convolution operation with kernel size of $C_2 \times C_1 \times 1 \times 1$ and $C_1 \times C_2 \times 1 \times 1$, respectively. $W^{\text{reduce}} \in R^{C_2 \times C_1}$ and $W^{\text{restore}} \in R^{C_1 \times C_2}$ are the trainable parameters of $\mathcal{F}_1(\cdot)$ and $\mathcal{F}_2(\cdot)$, respectively. C_1 and C_2 are the channel numbers of $\boldsymbol{H}^{\text{input}}$ and $\boldsymbol{H}^{\text{reduce}}$, respectively. The $\psi(\cdot)$ denotes the spatial attention kernel generation mechanism. $\delta(\cdot)$ and $\Theta(\cdot)$ denote the rectified linear unit (ReLU) activation function and batch normalization, respectively.

D. Residual Spatial Attention Kernel Generation Network

Based on the RSAB, we design an RSAKGN which consists of a 1×1 convolutional layer, three RSABs, and a fully connected classification layer. The first 1×1 convolutional layer with a kernel size of $C_1 \times C \times 1 \times 1$ is used to map the input HSI data into a compact feature space to reduce the computational complexity of the subsequent FE process. The three RSABs are stacked to extract discriminative deep semantic features from the compact feature space which generated by the first 1×1 convolutional layer for HSIC. The fully connected classification layer, which consists of a GAP, a fully connected layer, and a softmax classifier, is responsible to generate the predicted classification labels. In this article, C_1 is set to 96, 64, and 96 on the Indian Pines (IN), University of Pavia (UP), and Kennedy Space Center (KSC) datasets, respectively. C_2 is set to 32 on all three datasets.

IV. Experiments

A. Datasets Description

To verify the effectiveness of the proposed model, we conduct experiments on three publicly available hyperspectral datasets: IN, UP, and KSC.

The IN dataset was imaged by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992 on the Indiana Pines test field in IN, USA. It consists of 145×145 pixels with 220 spectra bands range from 0.4 to 2.5 μm. The spectral resolution is 10 nm and the spatial resolution is 20 m/pixel. Its ground truth contains 16 land-cover types with 10 249 labeled pixels. After removing 20 bands that do not reflect water and zero-bands, the remaining 200 bands are used for classification.

The UP data are captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. It consists of

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TABLE I
SAMPLE NUMBERS IN TRAINING, VALIDATION, AND TESTING SETS OF IN DATASET

Class	Name	Train	Val	Test
1	Alfalfa	2	4	40
2	Corn-notill	27	54	1347
3	Corn-mintill	19	35	776
4	Corn	4	9	224
5	Grass-pasture	9	18	456
6	Grass-tress	14	28	688
7	Grass-pasture-mowed	2	5	21
8	Hay-windrowed	10	19	449
9	Oats	3	4	13
10	Soybean-notill	24	41	907
11	Soybean-mintill	41	88	2326
12	Soybean-clean	14	26	553
13	Wheat	4	8	193
14	Woods	18	42	1205
15	Buildings-Grass-Tress-Drives	7	15	364
16	Stone-Steel-Towers	2	4	87
	Total	200	400	9649

TABLE II
SAMPLE NUMBERS IN TRAINING, VALIDATION, AND TESTING SETS OF UP DATASET

Class	Name	Train	Val	Test
1	Asphalt	31	62	6538
2	Meadows	88	174	18387
3	Gravel	10	20	2069
4	Trees	13	26	3025
5	Painted metal sheets	8	16	1321
6	Bare Soil	23	46	4960
7	Bitumen	9	18	1303
8	Self-Blocking Bricks	14	28	3640
9	Shadows	4	10	933
	Total	200	400	42176

TABLE III
SAMPLE NUMBERS IN TRAINING, VALIDATION, AND TESTING SETS OF KSC DATASET

Class	Name	Train	Val	Test
1	Scrub	29	58	674
2	Willow swamp	9	19	215
3	CP hammock	10	20	226
4	Slash pine	10	19	223
5	Oka/Broadleaf	6	12	143
6	Hardwood	9	18	202
7	Swamp	4	8	93
8	Graminoid marsh	17	33	381
9	Spartina marsh	20	40	460
10	Cattail marsh	16	31	357
11	Salt marsh	16	32	371
12	Mud flats	19	39	445
13	Water	35	71	821
	Total	200	400	4611

610 × 340 pixels with 115 spectral bands range from 0.43 to 0.86 μm. The spectral resolution is 4 nm and the spatial resolution is 1.3 m/pixel. Its ground truth contains nine land-cover types with a total of 42 776 labeled pixels. After removing 12 noise-affected bands, a total of 103 spectral bands are used for classification.

The KSC dataset was acquired by the AVIRIS sensors. It consists of 512 × 614 pixels with 224 spectral bands range from 0.4 to 2.5 μm. The spectral resolution is 10 nm and the spatial resolution is 18 m/pixel. Its ground truth contains 13 object categories with a total of 5211 labeled pixels. After removing water absorption bands and low signal-to-noise ratio bands, the remaining 176 spectral bands were used for classification.

We randomly sample each hyperspectral dataset according to a certain proportion, 200 labeled samples are used as the training set, 400 labeled samples as the validation set, and the remaining labeled samples as the testing set. For the IN dataset, we ensure that each category in the training set and validation set contains at least one labeled sample to avoid the situation where there are no training samples for rare categories. Tables I–III list the number of samples in the training, validation, and test sets of the three HSI datasets, respectively.

B. Experimental Configuration

All experiments are performed with PyTorch deep learning framework on a computer with a AMD Ryzen 3700× CPU with 64-GB RAM, a NVIDIA GeForce GTX1060 GPU with 6-GB RAM and Windows 10 operating system. We use He normal distribution initialization method [49] to initialize the parameters of all CNNs. All learnable parameters are optimized by the Adam optimizer [50]. The initial learning rate for model training was set to 0.002 and decreased by 5% every 30 epochs. The batch size and training epochs are set to 50 and 200, respectively.

In our experiments, we adopt the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) to quantitatively evaluate the classification performance of the algorithm. We run ten times for all experiments using randomly sampled training samples and report the mean and standard deviation of the three evaluation metrics above.

C. Parameters Setting

1) Patch Size: To study the effect of patch size S on the performance of RSAKGN, we performed experiment with

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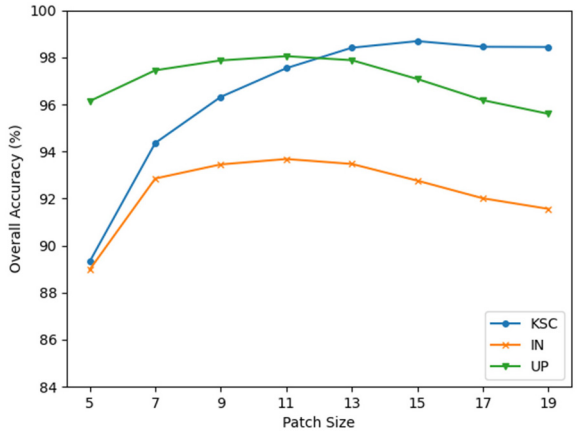


Fig. 7. OA (%) of RSAKGN with different patch sizes on IN, UP, and KSC datasets.

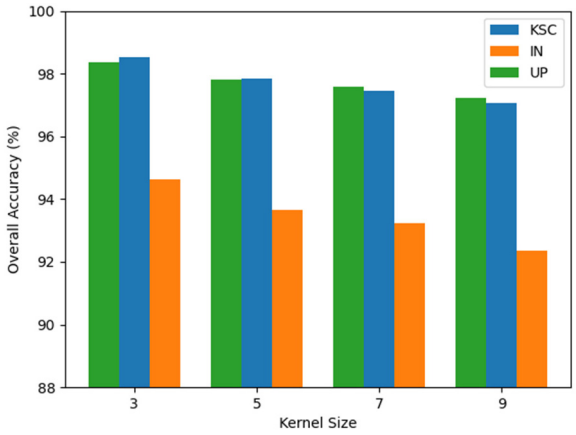


Fig. 8. OA (%) of RSAKGN with different kernel sizes on IN, UP, and KSC datasets.

different patch sizes {5, 7, 9, 11, 13, 15, 17} to find the optimal patch size. The results are shown in Fig. 7. For IN and UP datasets, with the increase of patch size, the OA first increases and then decreases, and when the patch size is set to 11×11, the OA reaches the maximum value. For the KSC dataset, with the increase of the patch size, the OA continues to increase, when the spatial size reaches to 13×13, the OA begins to converge. Considering that a larger patch size will increase the computational cost, the patch sizes of IN, UP, and KSC datasets are set to 11×11, 11×11, and 13×13, respectively.

2) Kernel Size: The kernel size L determines the receptive field of the spatial attention kernel, which is one of the important factors affecting the classification performance. We experiment with different kernel sizes {3, 5, 7, 9} to find the optimal kernel size.

As shown in Fig. 8, for all three datasets, the OA continued to decrease with the increase of kernel size and reached a maximum value when the kernel size is set to 3×3. Contrary to common sense, increasing the receptive field does not improve classification performance but reduces OA. We consider that RSAKGN dynamically generates its own unique spatial attention kernel at different spatial locations,

TABLE IV
OA (%) OF RSAKGN WITH DIFFERENT GROUP NUMBERS ON IN, UP, AND KSC DATASETS

Group Number	IN	UP	KSC
2	94.14±0.41	98.01±0.52	98.04±0.41
4	94.29±0.35	98.13±0.31	98.26±0.45
8	**94.82±0.52**	98.24±0.29	98.40±0.38
16	94.46±0.55	**98.36±0.32**	**98.57±0.37**
32	93.83±0.63	98.21±0.42	98.25±0.48

TABLE V
OA (%) OF RSAKGN WITH DIFFERENT FEATURE REPRESENTATIONS ON IN, UP, AND KSC DATASETS

Feature Eigenvector	IN	UP	KSC
ident	94.69±0.88	98.29±0.48	98.38±0.45
avg	94.48±1.12	98.13±0.46	98.15±0.47
max	94.29±1.07	98.09±0.41	98.06±0.47
avg + *max*	94.62±0.73	98.21±0.32	98.23±0.28
ident + *avg*	94.74±0.65	98.30±0.39	98.43±0.46
ident + *max*	94.64±0.71	98.26±0.32	98.36±0.32
ident + *avg* + *max*	**94.82±0.47**	**98.48±0.32**	**98.57±0.37**

it only focuses on the contextual information in the current spatial neighborhood. Therefore, an excessively large receptive field will introduce too much irrelevant information from other locations, which will have a negative impact on the spatial correlation calculation. Finally, the kernel size of IN, UP, and KSC datasets are all set to 3×3.

3) Number of Groups: The number of groups G is one of the important factors affecting the classification performance of the RSAKGN, which determines the number of spatial attention kernels in the model. We performed experiment on a set of different group numbers {2, 4, 8, 16, 32} to find the optimum group number G. The results are shown in Table IV. When the number of groups G is set to 8, 16, and 16, the proposed RSAKGN obtains the maximum OA value on IN, UP, and KSC datasets, respectively. For the UP and KSC datasets, which contain more detailed scenes, a larger number of groups can help the model to obtain more discriminative semantic features, thus helping to improve the classification accuracy. For the IN dataset containing a large smooth area, the excessive number of groups will introduce redundant features, resulting in a reduction of the feature representation capability, thus reducing the classification accuracy.

D. Ablation Experiment

In this section, we performed ablation experiments on three datasets to verify the effectiveness of the proposed modules and their contribution to the model.

First, to explore the impact of different feature representations on model performance, various combinations of those three eigenvectors are utilized to conduct experiments. The experimental results are presented in Table V,

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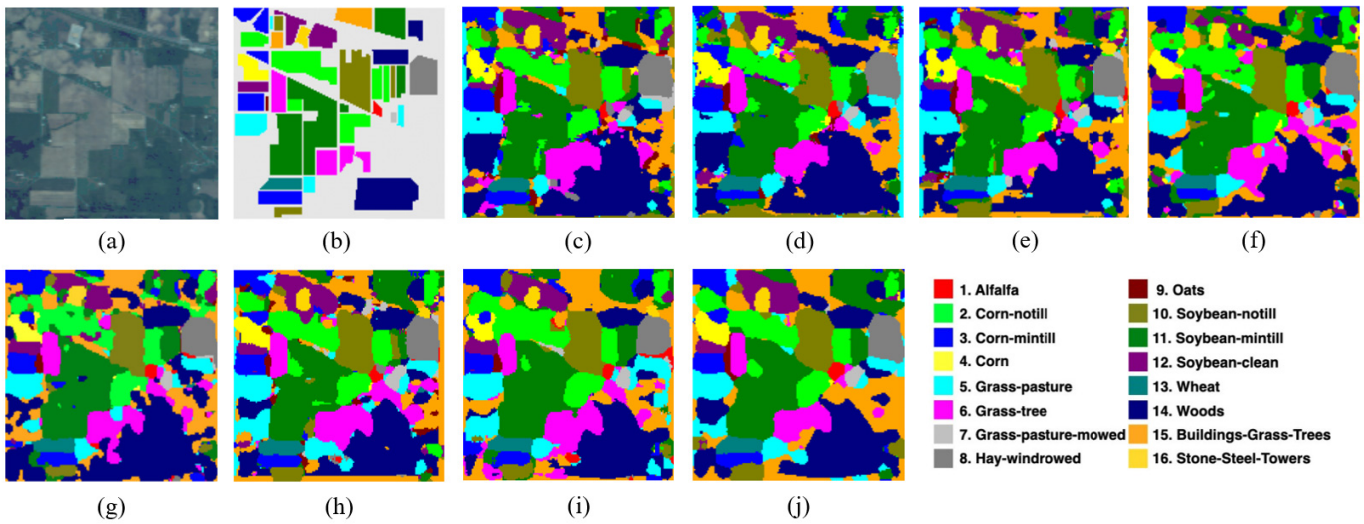


Fig. 9. Classification maps of the IN dataset performed by different HSIC methods: (a) false-color map; (b) ground truth map; and (c)–(j) DRN, MSRN, RSSAN, SSRN, SSAN, DRIN, SSTN, and RSAKGN, respectively.

TABLE VI
OA (%) PERFORMED BY DIFFERENT FEATURE FILTERS ON IN, UP, AND KSC DATASETS

Spatial Kernel	IN	UP	KSC
Convolution	87.21±1.08	95.67±1.05	95.58±0.68
Spatial Attention	92.16±0.91	96.58±0.77	96.51±1.42
Involution	92.99±0.83	97.38±0.63	96.78±1.02
SAKGM	**94.23±0.81**	**98.10±0.47**	**98.02±0.62**

TABLE VII
OA (%) OF RSAKGN WITH DIFFERENT MODULES ON IN, UP, AND KSC DATASETS

Module	IN	UP	KSC
CNN	87.21±1.08	95.67±1.05	95.58±0.68
SAKGM	94.23±0.81	98.10±0.47	98.02±0.62
RSAB	**94.82±0.47**	**98.36±0.32**	**98.57±0.37**

where the ident, avg, and max denote the $\boldsymbol{d}_j^{\text{ident}}$, $\boldsymbol{d}_j^{\text{avg}}$, and $\boldsymbol{d}_j^{\text{max}}$, respectively. For the combination of single eigenvector, all three eigenvectors provided satisfactory classification results. $\boldsymbol{d}_j^{\text{ident}}$ achieved the best OA due to its abundant discriminative spectral signature, and $\boldsymbol{d}_j^{\text{avg}}$ and $\boldsymbol{d}_j^{\text{max}}$ achieved relatively lower OA than $\boldsymbol{d}_j^{\text{ident}}$ due to the focus on global features representation. After combing eigenvectors in pairs, OAs of all combinations are improved compared to the single eigenvector and a smaller standard deviation of OA is obtained. When all three eigenvectors are combined as the feature representation, model achieved the best classification performance and obtained a smaller standard deviation of OA. The experimental results demonstrated that the discriminative spectral signatures provided by the local identity eigenvector are more useful for feature representation and the global statistics can provide an effectively complement for feature representation. In addition, the combination of various eigenvectors for feature representation can improve the classification performance of model and make model more robust to the training process.

To verify the effectiveness of the proposed module, the SAKGM is compared with the ordinary convolution module, spatial attention module, and involution kernels module. The experimental results are shown in Table VI. As shown in Table VI, the proposed SAKGM achieved the best classification results because it can fully exploit the diverse context information over different locations and dynamically generate the spatial attention kernels, which has stronger spatial correlations than other kernels.

Ablation experiments are conducted on three datasets to verify the effectiveness of the proposed modules and their contribution to the model. We take the ordinary CNN (i.e., only utilizes convolution kernels to extract sematic features without any other modules) as the baseline and gradually add various modules to study their role in the model, and the experimental results are shown in Table VII. The ordinary CNN is unable to achieve satisfactory classification results. After adding SAKGM, the classification performance of the model has been significantly improved, which demonstrates that SAKGM can effectively improve the classification performance of the model. Then, we combine SAKGM with residual learning framework to obtain the RSAB and construct the complete RSAKGN; the classification performance of the model is further improved which proves the effectiveness of the residual learning mechanism.

The ablation experimental results exhibit that both modules have a positive effect on improving the classification performance, and can be organically combined to jointly improve the performance of the model.

E. Comparison With the State-of-the-Arts

In this section, we compare the proposed RSAKGN with seven deep learning based methods: a deep residual network

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TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH 200 TRAINING SAMPLES ON IN DATASET

Class	DRN	MSRN	RSSAN	SSRN	SSAN	DRIN	SSTN	RSAKGN
1	42.75	60.00	55.12	71.03	95.00	80.75	72.50	**95.75**
2	79.99	77.15	84.71	85.99	80.12	86.99	**92.05**	91.62
3	85.35	78.02	80.93	91.61	85.79	87.41	87.86	**91.04**
4	82.52	86.56	81.25	91.83	89.95	95.80	94.87	**97.37**
5	75.88	83.14	90.57	89.63	89.58	91.60	85.46	**91.27**
6	94.07	95.76	96.66	96.38	90.80	97.06	97.18	**97.89**
7	98.11	91.43	90.47	**99.05**	**100.00**	98.57	**100.00**	**100.00**
8	99.58	99.80	99.11	99.89	95.91	99.40	**100.00**	**100.00**
9	**100.00**	**97.69**	**100.00**	99.23	**100.00**	97.69	**100.00**	97.69
10	87.22	84.98	90.08	89.08	86.23	90.26	**92.66**	89.51
11	90.02	84.43	92.09	96.05	91.75	95.41	**97.41**	96.96
12	69.04	65.70	87.34	83.62	79.11	82.68	87.80	**88.35**
13	96.99	96.79	99.48	96.84	98.94	96.74	**100.00**	99.22
14	97.06	98.22	97.76	99.25	97.80	99.37	98.81	**99.30**
15	79.67	85.02	85.44	93.16	91.29	95.11	82.58	**96.87**
16	**99.20**	97.13	97.88	99.20	91.76	98.51	95.29	96.58
OA(%)	87.21±1.08	85.43±1.11	90.54±2.71	91.97±1.18	89.74±1.52	92.99±0.83	93.79±0.72	**94.82±0.47**
AA(%)	86.07±1.54	86.36±1.50	89.31±3.32	92.63±1.58	91.50±2.03	93.33±1.27	92.78±0.93	**95.59±0.65**
κ×100	85.41±1.23	83.43±1.24	89.27±3.12	90.89±1.35	88.32±1.72	92.01±0.92	92.88±0.78	**94.09±0.52**
Training time (s)	22.38±0.45	29.92±0.41	23.87±0.42	214.92±0.92	164.32±0.41	39.27±0.26	33.06±0.37	52.99±0.43
Testing time (s)	2.52±0.02	2.67±0.05	2.61±0.01	7.84±0.03	5.91±0.04	2.67±0.02	2.72±0.02	2.84±0.02

TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH 200 TRAINING SAMPLES ON UP DATASET

Class	DRN	MSRN	RSSAN	SSRN	SSAN	DRIN	SSTN	RSAKGN
1	94.14	97.51	95.12	97.51	96.95	95.65	97.56	**98.53**
2	99.54	99.60	99.45	99.66	99.25	99.56	**99.87**	99.70
3	89.90	95.10	91.20	94.09	78.58	93.15	93.03	**95.86**
4	**97.23**	88.55	92.26	89.28	94.03	95.32	92.81	93.71
5	99.32	93.53	99.62	99.76	99.24	99.28	**100.00**	**100.00**
6	94.52	92.77	93.73	95.54	98.35	96.93	98.13	**98.89**
7	80.81	99.30	91.79	99.15	96.42	96.67	99.52	**100.00**
8	86.18	92.95	92.12	91.48	94.16	93.72	92.99	**94.61**
9	97.12	97.07	96.46	**98.79**	95.57	97.76	94.89	96.38
OA(%)	95.67±1.05	96.50±0.47	96.25±0.93	97.08±0.61	96.79±0.64	97.38±0.63	97.79±0.49	**98.48±0.32**
AA(%)	93.21±1.56	95.15±0.62	94.64±1.15	96.14±0.95	94.73±1.09	96.49±1.01	96.53±0.82	**97.52±0.45**
κ×100	94.27±1.19	95.38±0.62	95.02±0.96	96.38±0.82	94.75±0.85	96.53±0.75	97.07±0.56	**97.75±0.36**
Training time (s)	16.29±0.33	33.50±0.47	16.61±0.34	110.09±0.25	95.20±1.33	27.53±0.41	23.28±0.28	55.50±0.31
Testing time (s)	6.68±0.08	7.48±0.06	7.04±0.05	17.61±0.03	15.15±0.18	7.54±0.04	7.32±0.11	9.09±0.03

(DRN) [29], SSRN [35], multiscale residual network (MSRN) [36], RSSAN [38], SSAN [39], SSTN [43], and DRIN [46]. All these methods take 3-D HSI cube as input data. To ensure the fairness of the experiments, both the proposed method and compared methods adopt the same experimental setup.

Tables VIII–X present the quantitative experimental results, including OA of each classes, OA, AA, and Kappa coefficients for all classes, performed by different methods on three datasets, respectively. As shown in Tables VIII–X, the OA of all CNN-based models, including DRN, MSRN, RSSAN, and SSAN, is significantly lower than that of non-CNN-structured models, like SSTN, DRIN, and RSAKGN, for all three datasets. The DRIN, utilizing dynamic involution kernels for FE, achieved relatively good classification results. The SSTN, performing attention mechanism for semantic FE, achieved the second highest OA with limited number of samples. But SSTN only pays attention to the global feature information and ignores the diversity of local spatial interaction. The proposed RSAKGN achieved the highest OA on all three datasets. The OAs of RSAKGN on IN dataset, UP dataset, and KSC

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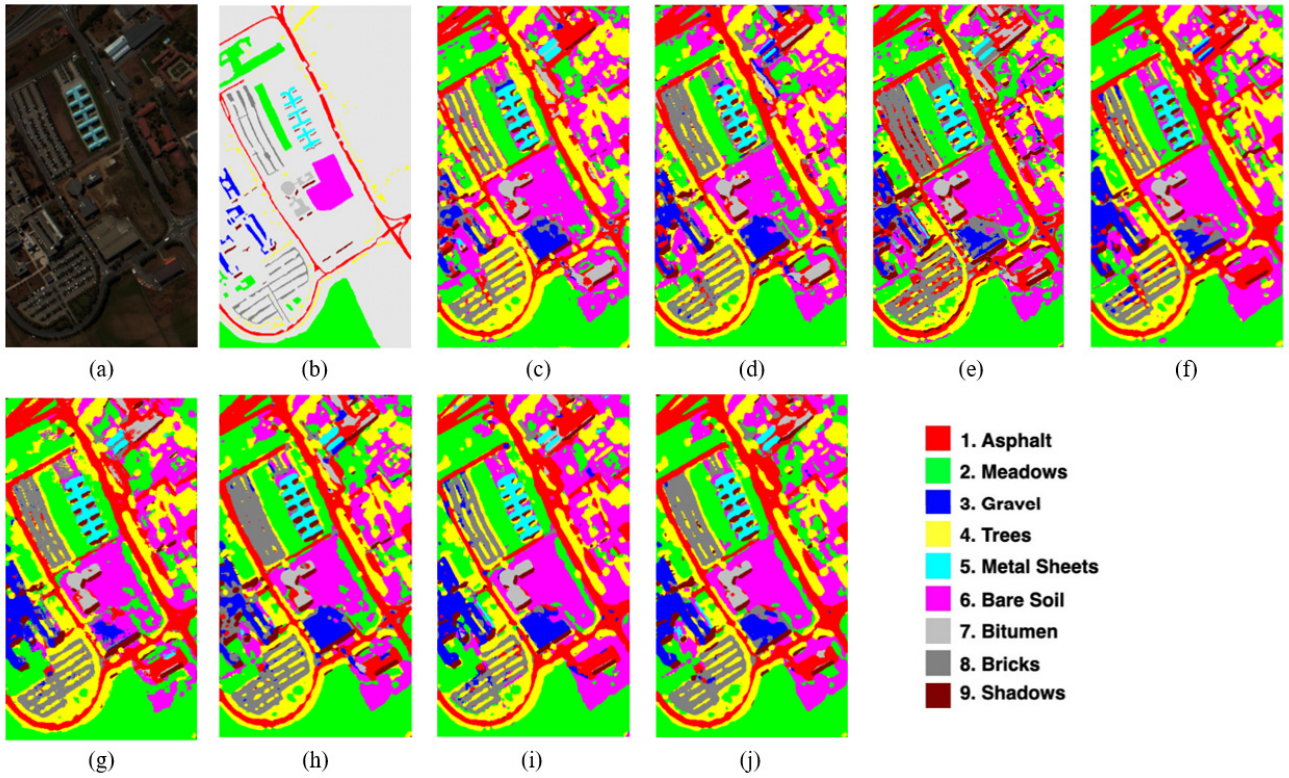


Fig. 10. Classification maps of the UP dataset performed by different HSIC methods: (a) false-color map; (b) ground truth map; and (c)–(j) DRN, MSRN, RSSAN, SSRN, SSAN, DRIN, SSTN, and RSAKGN, respectively.

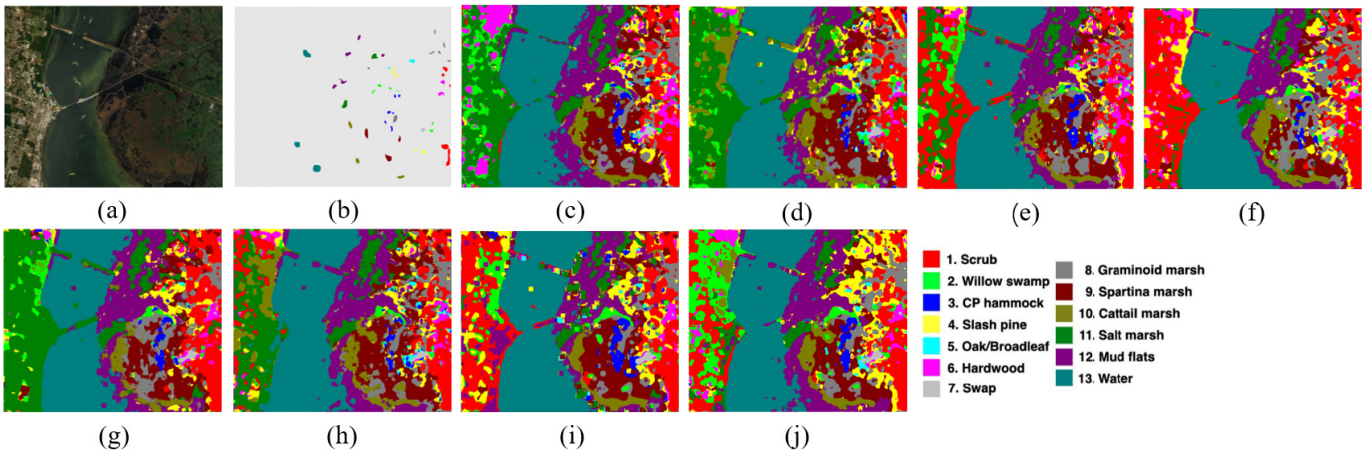


Fig. 11. Classification maps of the KSC dataset performed by different HSIC methods: (a) false-color map; (b) ground truth map; and (c)–(j) classification maps of DRN, MSRN, RSSAN, SSRN, SSAN, DRIN, SSTN, and RSAKGN, respectively.

dataset are 94.82%, 98.48%, and 98.57%, respectively, which improves 1.03%, 0.69%, and 1.27% OA of the SSTN.

Figs. 9–11 show the false-color map of HSI, the ground truth maps of HSI, and the classification maps generated by all HSIC methods on three datasets. Those qualitative classification maps are consistent with the quantitative experimental results reported in Tables VIII–X. The proposed RSAKGN can generate a more uniform and smooth classification map with fewer misclassified pixels. For datasets like the IN dataset that contains large smooth areas, RSAKGN can effectively maintain the consistency of pixel categories within the same segmentation block. For datasets like the UP and KSC datasets that contain more detailed scenes, RSAKGN can not only effectively reduce the probability of misclassification, but also more accurately describe the boundaries between different categories.

The training and testing times of different HSIC methods on three datasets are listed in the bottom of Tables VIII–X. As a result of utilizing 3-D CNN architecture, SSRN and SSAN need longer time for training and testing. Relying on the efficient calculation fashion of 2-D CNN architectures, DRN, MSRN, and RSSAN need less time for training and testing.

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TABLE X
CLASSIFICATION RESULTS OF DIFFERENT METHOD WITH 200 TRAINING SAMPLES ON KSC DATASET

Class	DRN	MSRN	RSSAN	SSRN	SSAN	DRIN	SSTN	RSAKGN
1	97.18	99.78	99.70	**100.00**	99.94	**100.00**	**100.00**	99.72
2	81.39	84.79	85.12	80.94	89.21	86.04	93.72	**98.78**
3	78.76	98.09	98.23	99.07	97.08	94.69	99.07	**100.00**
4	92.38	78.48	88.34	**99.07**	87.44	89.69	80.19	93.32
5	70.35	68.60	56.64	**99.27**	62.10	70.76	81.02	75.11
6	94.06	99.11	96.53	88.08	93.86	99.65	94.82	**99.41**
7	99.74	97.74	**100.00**	**100.00**	99.14	99.25	89.89	99.57
8	98.26	99.74	99.74	92.05	98.45	99.58	99.45	**100.00**
9	**100.00**	98.41	96.52	**100.00**	91.76	94.13	99.32	98.15
10	98.75	99.02	98.74	96.87	**100.00**	99.43	**100.00**	99.83
11	99.65	**100.00**	**100.00**	**100.00**	99.84	**100.00**	**100.00**	**100.00**
12	**100.00**	97.93	**100.00**	91.53	98.54	98.87	97.19	99.93
13	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	95.58±0.68	96.61±1.21	96.41±1.52	96.74±1.12	96.17±1.07	96.78±1.02	97.30±1.15	**98.57±0.37**
AA(%)	93.12±1.05	93.98±1.74	93.81±2.36	95.91±1.59	93.64±1.69	94.78±1.57	94.97±2.01	**97.22±0.47**
κ×100	95.07±0.83	96.23±1.38	96.01±1.74	96.37±1.21	95.72±1.19	96.41±1.18	96.99±1.23	**98.41±0.41**
Training time(s)	27.51±0.22	36.83±0.43	26.51±0.67	266.22±1.65	204.86±0.85	39.62±0.21	36.94±0.69	80.97±0.91
Testing time(s)	1.14±0.01	1.22±0.02	1.19±0.02	4.59±0.03	3.46±0.04	1.25±0.01	1.23±0.02	1.42±0.02

TABLE XI
NUMBER OF PARAMETERS OF DIFFERENT HSIC METHODS ON IN, UP, AND KSC DATASETS

	DRN	MSRN	RSSAN	SSRN	SSAN	DRIN	SSTN	RSAKGN
Indian Pines	324,112	251,256	106,162	735,884	341,350	62,344	20,965	67,504
Pavia University	239,625	193,545	70,357	396,993	232,585	52,353	16,212	74,281
Kennedy Space	303,085	236,632	96,868	651,125	313,720	59,749	19,865	91,789

DRIN and SSTN require relatively little time for training with their unique network structure. Due to a large amount of matrix multiplications involved, the proposed RSAKGN requires a relatively long time for training. Although the training time of RSAKGN is not satisfactory, it is still far less than that required for SSAN and SSRN. Note that RSAKGN is just a litter longer than abovementioned calculating efficient models in terms of the testing time, which means RSAKGN hardly causes extra computational cost in inference stage and can achieve better classification performance. Table XI lists the number of parameters of different HSIC methods on three datasets. As reported in Table XI, compared with DRN, MSRN, RSSAM, SSRN, and SSAN which utilize ordinary convolution kernels for FE, RSAKGN achieves better classification with fewer learnable parameters. It confirms that the RSAKGN is a lightweight model with better performance. Compared with DRIN and SSTN, the proposed model has more learnable parameters, but it significantly achieves better classification results. To sum up this section, the proposed method can greatly save computing resource and provide remarkable classification performance.

In addition, to verify the effect of varying training sample sizes on classification performance, extensive experiments are conducted. We performed the above eight HSIC algorithms with varying training samples sizes which range from 200 to 400 with an interval of 50, on IN, UP, and KSC

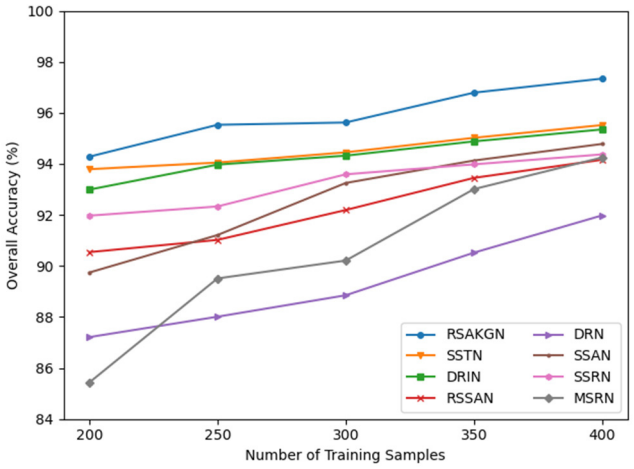


Fig. 12. OA (%) obtained by eight methods with different training sample sizes on the IN dataset.

datasets. As shown in Figs. 12–14, all models performed better with the increase of training samples. And those CNN-based models are sensitive to the training sample sizes, which also exhibiting that CNN-based models have a strong dependence on training sample size from another perspective. Compared with the reference methods, the proposed RSAKGN can consistently provide superior classification performance on three

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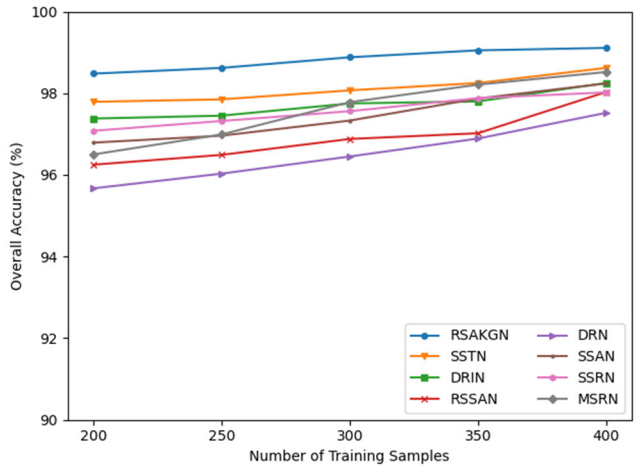


Fig. 13. OA (%) obtained by eight methods with different training sample sizes on the UP dataset.

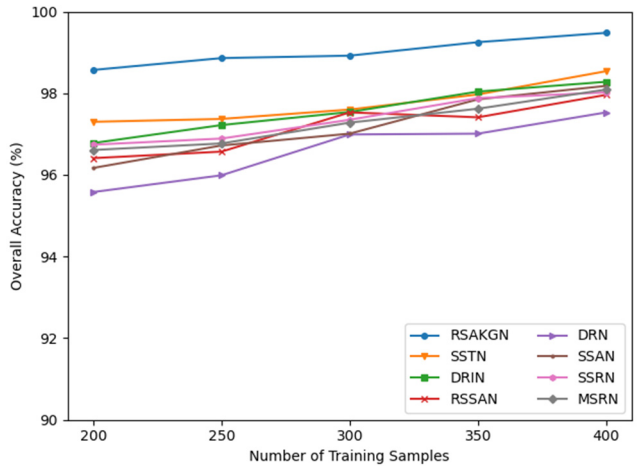


Fig. 14. OA (%) obtained by eight methods with different training sample sizes on the KSC dataset.

datasets. All these experimental results demonstrate that the RSAKGN can handle the small sample problem effectively and has a robustness for the training sample sizes.

V. Conclusion

In this article, a novel RSAKGN is proposed for HSIC, which has achieved the highest OA with limited training samples on three HSI datasets. First, a plug and play SAKGM is proposed, which can be flexibly embedded into models and replace the convolution kernel for semantic FE. The proposed SAKGM can dynamically generate spatial attention kernels according to the specific context information. To deepen the network structure and enhance the feature representation ability of deep semantic features, we combine the SAKGM with the residual learning mechanism to construct an RSAB. Finally, the RSAKGN is constructed by stacking several RSABs to achieve multilevel extraction of deep semantic features. The experimental results demonstrate the effectiveness of the proposed RSAKGN for coping with small sample size problems.

However, the SAKGM in RSAKGN is computationally intensive, which makes the training time of RSAKGN much longer than that of SSTN and DRIN. In the future, we will explore a concise and effective spatial kernel generation mechanism for HSIC.

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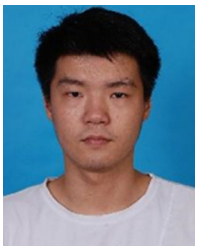
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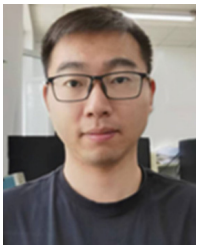
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